

1. Rearrange the terms in the alternating harmonic series such that its sum is $s \in \mathbb{R}$.

According to Riemann Theorem, a conditionally convergent series can be rearranged to converge to any value or diverge to $\pm\infty$.

The alternating harmonic series is $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$.

$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges to $\ln 2$, but $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{n} \right| =$

$= \sum_{n=1}^{\infty} \frac{1}{n}$ diverges $\Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ is conditionally convergent (harmonic series)

Rearranging the sum (one possible way): take one positive term, 2 negative terms and so on...

$$\begin{aligned} \Rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} &= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \dots = \ln 2 \\ &= \underbrace{1 - \frac{1}{2}}_{\frac{1}{2}} + \underbrace{\frac{1}{3} - \frac{1}{4}}_{\frac{1}{12}} + \underbrace{\frac{1}{5} - \frac{1}{6}}_{\frac{1}{30}} + \dots \\ &= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \dots = \\ &= \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right) = \frac{1}{2} \ln 2 \in \mathbb{R} \end{aligned}$$

Given a target $s \in \mathbb{R}$:

- start adding positive terms until the partial sum first exceeds s
- then add enough negative terms to make the partial sum go below s
- repeat the process

• Because the terms get smaller and smaller (as n increases), each jump above or below s becomes very small, so the partial sums settle down and approach s .

2. Let (F_m) be the Fibonacci sequence, with $F_0 = F_1 = 1$ and $F_m = F_{m-1} + F_{m-2}$. Study the convergence and find the value of the series $\sum_{m \geq 0} F_m x^m$.

$$\text{Let } G(x) = \sum_{m \geq 0} F_m x^m$$

$$G(x) = F_0 + F_1 x + F_2 x^2 + F_3 x^3 + \dots$$

$$\Rightarrow x \cdot G(x) = F_0 x + F_1 x^2 + F_2 x^3 + F_3 x^4 + \dots$$

$$\Rightarrow x^2 \cdot G(x) = F_0 x^2 + F_1 x^3 + F_2 x^4 + F_3 x^5 + \dots$$

$$\bullet G(x) = F_0 + F_1 x + F_2 x^2 + F_3 x^3 + \dots$$

$$\begin{aligned} F_2 x^2 + F_3 x^3 + \dots &= G(x) - F_0 - F_1 x \\ &= G(x) - 1 - x \quad \text{I} \end{aligned}$$

$$\bullet F_m = F_{m-1} + F_{m-2}$$

$$\Rightarrow \begin{cases} F_2 x^2 = (F_1 + F_0) x^2 = F_1 x^2 + F_0 x^2 \\ F_3 x^3 = (F_2 + F_1) x^3 = F_2 x^3 + F_1 x^3 \\ F_4 x^4 = (F_3 + F_2) x^4 = F_3 x^4 + F_2 x^4 \\ \dots \end{cases}$$

$$\begin{aligned} \Rightarrow F_2 x^2 + F_3 x^3 + F_4 x^4 &= (F_1 x^2 + F_0 x^2) + (F_2 x^3 + F_1 x^3) + (F_3 x^4 + F_2 x^4) + \dots = \\ &= \underbrace{F_1 x^2 + F_2 x^3 + F_3 x^4 + \dots}_{x \cdot G(x) - x} + \underbrace{F_0 x^2 + F_1 x^3 + F_2 x^4 + \dots}_{x^2 \cdot G(x)} = \end{aligned}$$

$$= x \cdot G(x) + x^2 \cdot G(x) - x \quad \text{II}$$

$$\text{From I and II} \Rightarrow G(x) - 1 - x = x \cdot G(x) + x^2 \cdot G(x) - x$$

$$G(x) - x \cdot G(x) - x^2 \cdot G(x) = 1 + x - x$$

$$G(x) (1 - x - x^2) = 1$$

$$G(x) = \frac{1}{1 - x - x^2} \quad \leftarrow \text{well defined except where } 1 - x - x^2 = 0, \text{ that means } x = \frac{-1 \pm \sqrt{5}}{2}$$

$$x_1 = \frac{-1+\sqrt{5}}{2} \approx 0,618$$

$$x_2 = \frac{-1-\sqrt{5}}{2} \approx -1,618$$

$$R = \max(|x_1|, |x_2|) \approx 0,618$$

$\Rightarrow$ the series converges (and its sum is $\frac{1}{1-x-x^2}$)
when $|x| < 0,618$ (that is $x \in (-0,618, 0,618)$)

3. Let C_m be the number of full binary trees with $m+1$ leaves (known as the Catalan number).

a) Find a recurrence relation for C_m .

C_m = the no. of full binary trees with $m+1$ leaves
 $C_0 = 1$ (single leaf)

For $m \geq 1 \Rightarrow$ the root must have 2 subtrees, left and right and each one is a full binary tree

Take $k+1$ leaves for the left subtree $\Rightarrow$

$\Rightarrow$ the right subtree has $(m+1)-(k+1) = m+1-k-1 =$
 $= m-k$ leaves

$$(0 \leq k \leq m-1)$$

For the left subtree, we count by C_k (because it has $k+1$ leaves)

For the right subtree, we count by C_{m-k-1} (because it has $m-k$ leaves)

For each k , the number of such trees is $C_k \cdot C_{m-k-1}$

$$\Rightarrow \begin{cases} C_m = \sum_{k=0}^{m-1} C_k \cdot C_{m-k-1}, & \text{for } m \geq 1 \\ C_0 = 1 \end{cases}$$